CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · PYQ

Chapter 19: Constructions (Circles)

Previous year board questions, topic-wise: 6 questions

The tag after each question gives the board examination year and the marks. Solutions are in a separate PDF with the same question numbers.

1 Circumscribed circle of a triangle

1. Using ruler and compasses only, construct $\triangle ABC$ with $AB = 6$ cm, $AC = 4.5$ cm and $\angle BAC = 120^\circ$. Construct the circle that passes through A, B and C, and measure its radius. ICSE 2025, 4 marks

2. Using ruler and compasses, construct $\triangle ABC$ with $AB = 6$ cm, $\angle BAC = 120^\circ$ and $AC = 5$ cm. Construct the circle passing through A, B and C, and measure its radius. ICSE 2023, 4 marks

3. Using ruler and compasses, construct $\triangle ABC$ with $AB = 3$ cm, $BC = 4$ cm and $\angle ABC = 90^\circ$. Construct its circumscribing circle and measure the radius. ICSE 2020, 4 marks

4. Using ruler and compasses only, construct $\triangle ABC$ with $BC = 5$ cm, $AB = 6.5$ cm and $\angle ABC = 120^\circ$. (a) Construct its circumcircle. (b) Construct the cyclic quadrilateral $ABCD$ in which D is equidistant from AB and BC. ICSE 2018, 4 marks

2 Tangents from an external point

5. Using ruler and compasses, draw a circle of radius 4 cm with centre O. Mark a point P 7 cm from O and construct the two tangents from P to the circle. Measure the length of a tangent. ICSE 2019, 4 marks

3 Inscribed circle of a triangle

6. Using ruler and compasses, construct $\triangle ABC$ with $AB = 7$ cm, $\angle CAB = 60^\circ$ and $AC = 5$ cm. Construct the locus of points equidistant from (a) AB and AC (b) BA and BC. Hence construct the circle that touches the three sides of the triangle internally. ICSE 2017, 4 marks

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